

6. PROJECT TESTING

FitBuddy – Testing & Local Deployment Verification

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Role: Testing & Local Deployment Verification

6.1 Introduction

FitBuddy was tested to verify user input validation, AI-based fitness-plan generation, feedback processing, FastAPI operations, database storage, and local deployment.

6.2 Testing Areas

- User input validation
- AI-based fitness-plan generation
- Different fitness goals
- FastAPI endpoint testing using Swagger UI
- Feedback processing and updated-plan generation
- Database storage and data handling
- Local deployment verification

6.3 Test Cases

Test Case	Input / Action	Expected Result	Actual Result	Status
TC01	Enter valid user details	Fitness plan generated	Fitness plan generated	Pass
TC02	Select different fitness goals	Plan generated according to selected fitness goal	Plan generated according to input	Pass
TC03	Leave required fields empty	Validation message displayed	Validation message displayed	Pass
TC04	Send valid API request through Swagger UI	AI response displayed	AI response displayed	Pass
TC05	Submit feedback for generated plan	Updated plan generated	Updated plan generated	Pass
TC06	Test FastAPI endpoints using Swagger UI	Endpoint responds correctly	Endpoint responded correctly	Pass
TC07	Check database storage	User/plan information stored	Information stored	Pass

Testing Status Note: The results in the table correspond to the application-testing evidence documented in Section 6.4.

6.4 Application Testing Evidence

The following screenshots provide application-level evidence for the corresponding testing activities performed in the local environment.

6.4.1 User Input Verification

The FitBuddy input page was checked to verify that the user can enter basic information such as name, age, weight, fitness goal, intensity, and preferences before generating a plan.

Figure 1 – FitBuddy User Input / 7-Day Fitness Plan Form

The screenshot shows a web browser window with the FitBuddy AI Fitness Plan Generator. The form is titled "Create Your 7-Day Fitness Plan" and includes fields for Name, Age, Weight (kg), Fitness Goal, Intensity, and Preferences / Limitations. A "Generate My Workout Plan" button is at the bottom. The browser's address bar shows "127.0.0.1:8000".

FitBuddy
AI Fitness Plan Generator

[View All Users](#)

Create Your 7-Day Fitness Plan

Enter your basic information and let Gemini create a personalized general wellness plan.

Name
Enter your name

Age
Enter age

Weight (kg)
Example: 60

Fitness Goal
Select a goal

Intensity
Select intensity

Preferences / Limitations
Example: I prefer home workouts. I have no equipment.

Note: FitBuddy provides general fitness and wellness information. It does not provide medical diagnosis or treatment.

[Generate My Workout Plan](#)

6.4.2 AI-Based 7-Day Workout Plan Generation

After valid input was submitted, FitBuddy displayed a generated 7-day workout plan. This provides evidence for the workout-plan generation test.

Figure 2 – Generated 7-Day Workout Plan

The screenshot shows the generated 7-day workout plan. It includes a "7-Day General Wellness Plan" section with a "Your 7-Day Workout Plan" table. The table lists exercises for Day 1, Day 2, and Day 3, including Warm-up, Bodyweight squats, and Arm Circles. The browser's address bar shows "127.0.0.1:8000/generate-workout".

FitBuddy
Your AI Fitness Plan

[Create New Plan](#)

New Plan

7-Day General Wellness Plan

Hello Test User!

A simple beginner-friendly home wellness plan with movement, recovery, and rest.

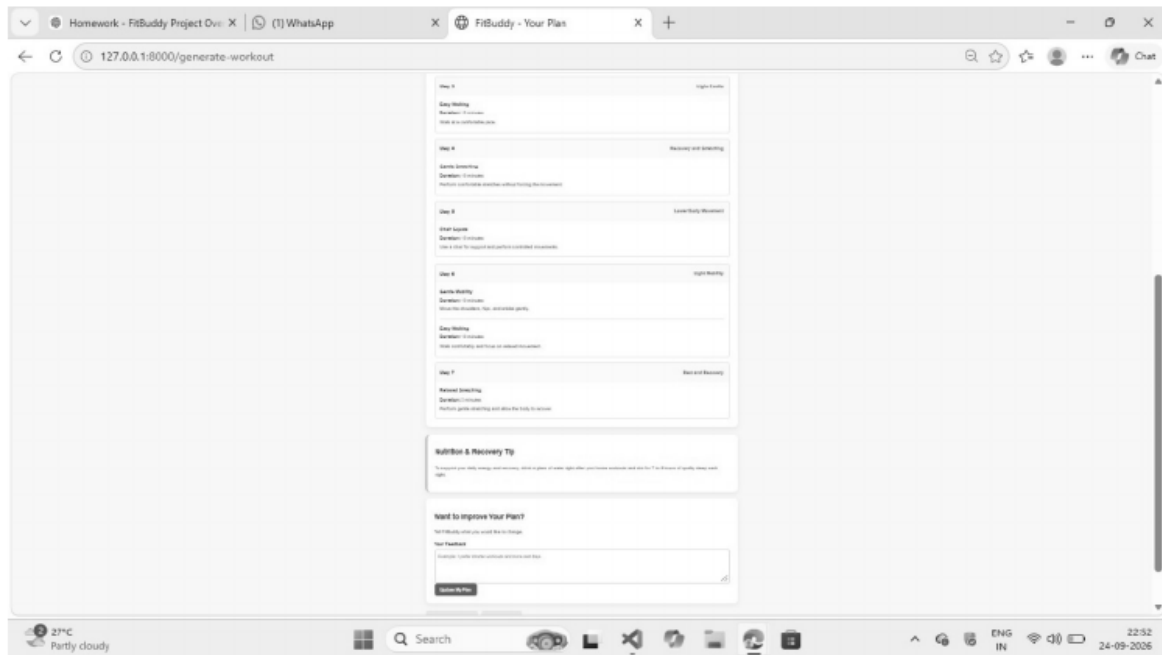
Your 7-Day Workout Plan

Day	Focus
Day 1	Full Body Movement
Warm-up Duration: 5 minutes Gentle walking and easy mobility movements.	
bodyweight squats Duration: 10 minutes Perform comfortable controlled squats with good posture.	
Day 2	Upper Body and Mobility
Arm Circles Duration: 5 minutes Move the arms gently in controlled circles.	
Wall Push-ups Duration: 10 minutes Use a wall for support and perform comfortable repetitions.	
Day 3	Light Cardio

6.4.3 Nutrition & Recovery Verification

The generated plan also displays a Nutrition & Recovery Tip, which was checked as part of the application output verification.

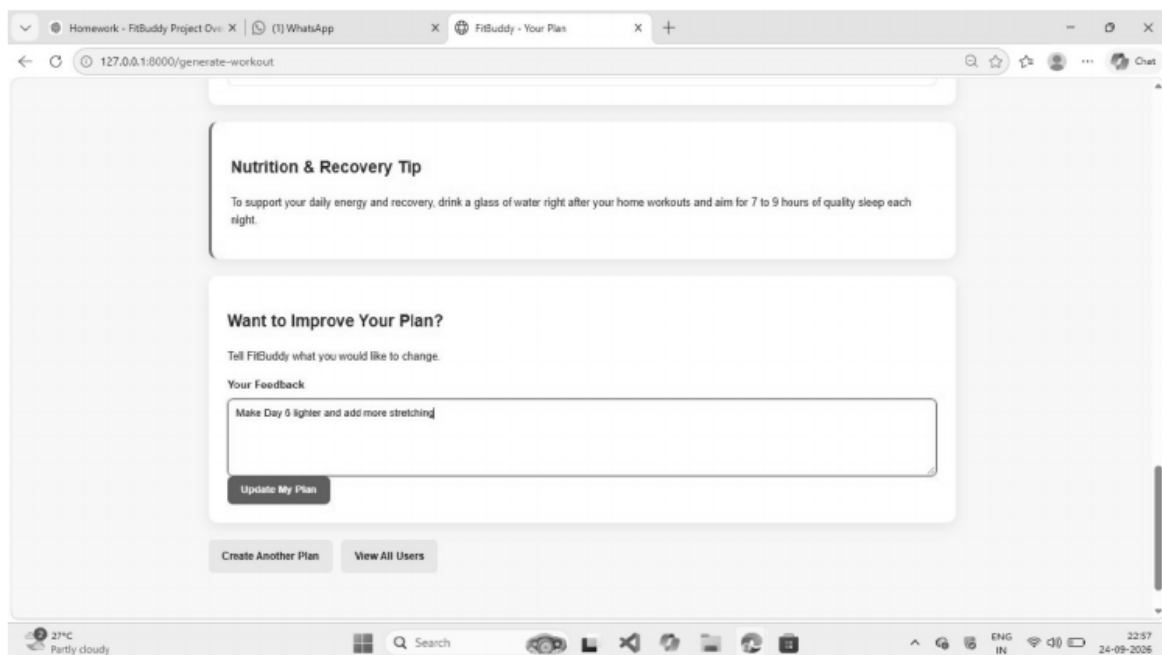
Figure 3 – Nutrition & Recovery Tip



6.4.4 Feedback Processing

The feedback section allows the user to enter a requested change to the generated plan. This was checked as part of the feedback-processing test.

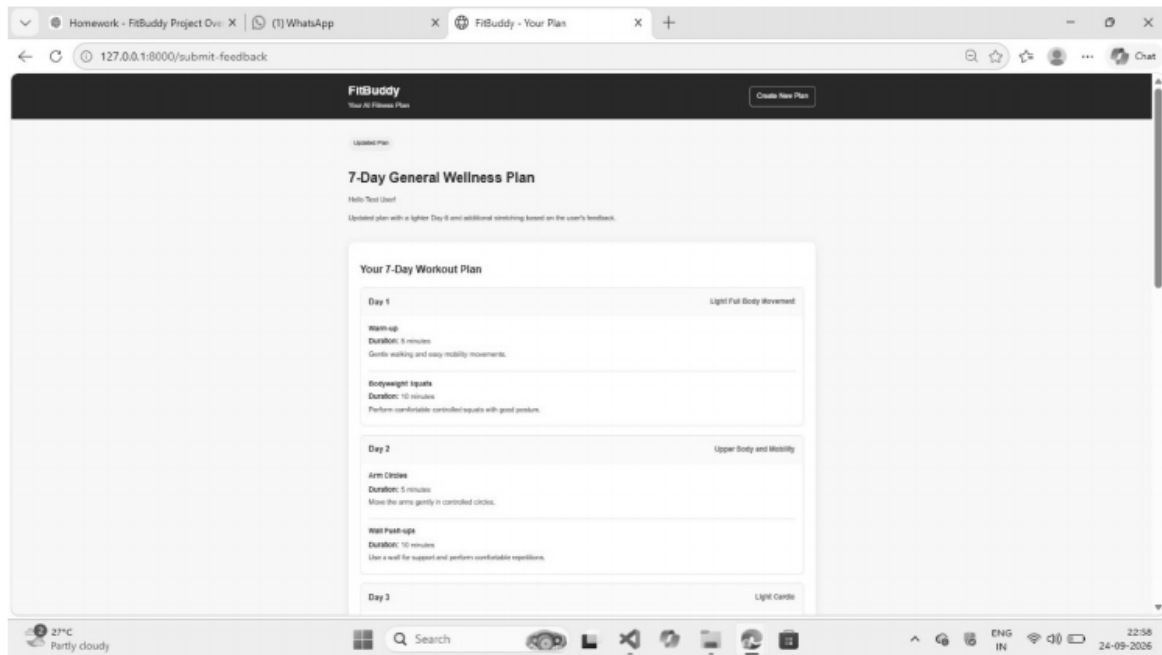
Figure 4 – Feedback Section



6.4.5 Updated Plan After Feedback

After feedback was submitted, FitBuddy displayed an updated 7-day plan reflecting the requested change. This provides evidence for the updated-plan generation test.

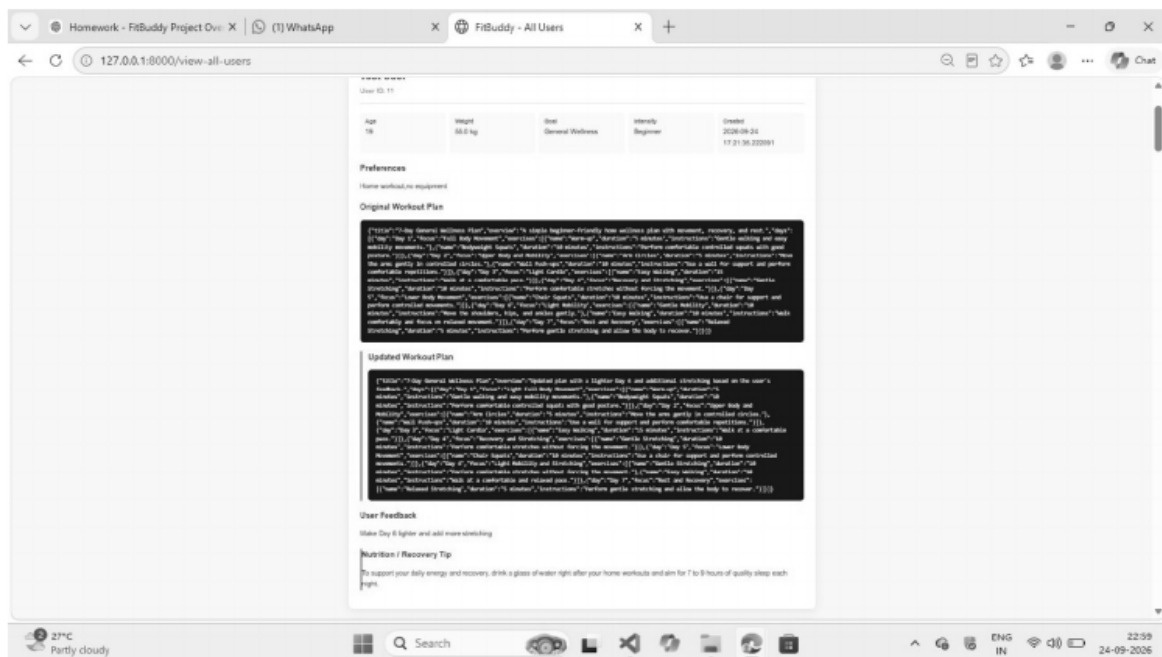
Figure 5 – Updated Plan After User Feedback



6.4.6 Database Storage Verification

The stored user information, original workout plan, updated workout plan, feedback, and Nutrition/Recovery Tip were viewed in the application's database view. This provides evidence for the data-storage test.

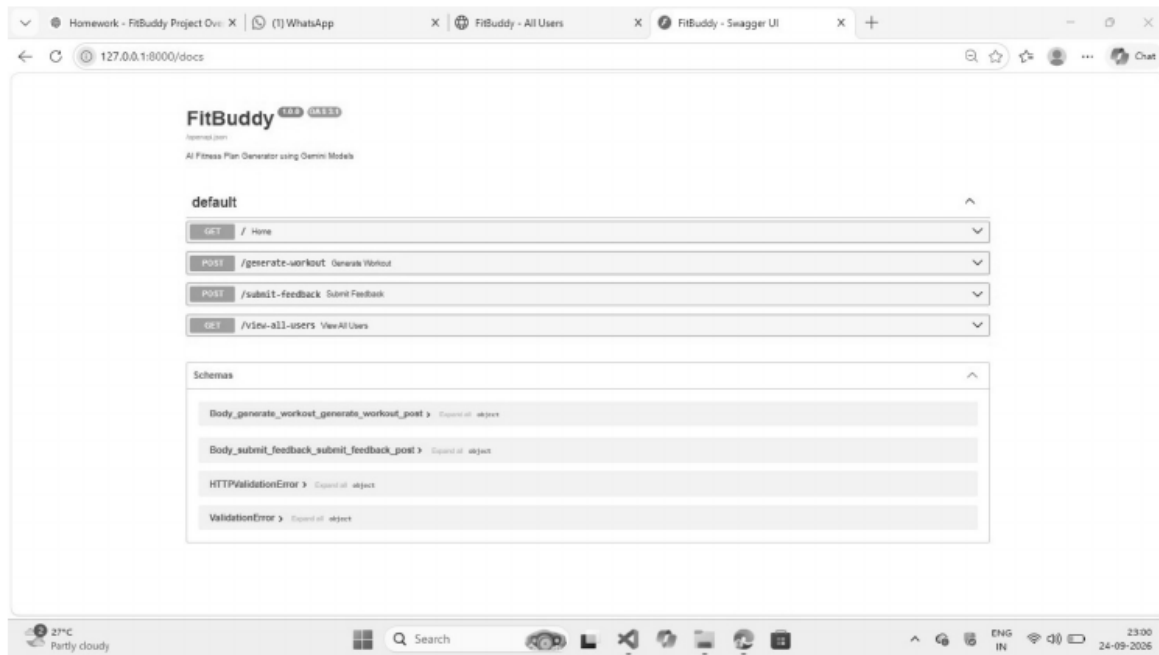
Figure 6 – Stored User and Workout Information



6.4.7 API Testing Using Swagger UI

The Swagger UI page was checked to verify the available FastAPI endpoints, including Generate Workout, Submit Feedback, and View All Users.

Figure 7 – Swagger UI API Endpoints



6.5 Local Deployment Verification

FitBuddy was executed in the local environment. The application pages, API endpoints, generated plans, feedback flow, updated plan, and stored data were checked during the verification process.

6.6 Bug Fixing – Optional

If an actual bug was identified and fixed during testing, it should be documented using the following format. Cause and solution should not be guessed.

Bug: _____

Cause: _____

Solution: _____

Testing After Fix: _____

Status: _____

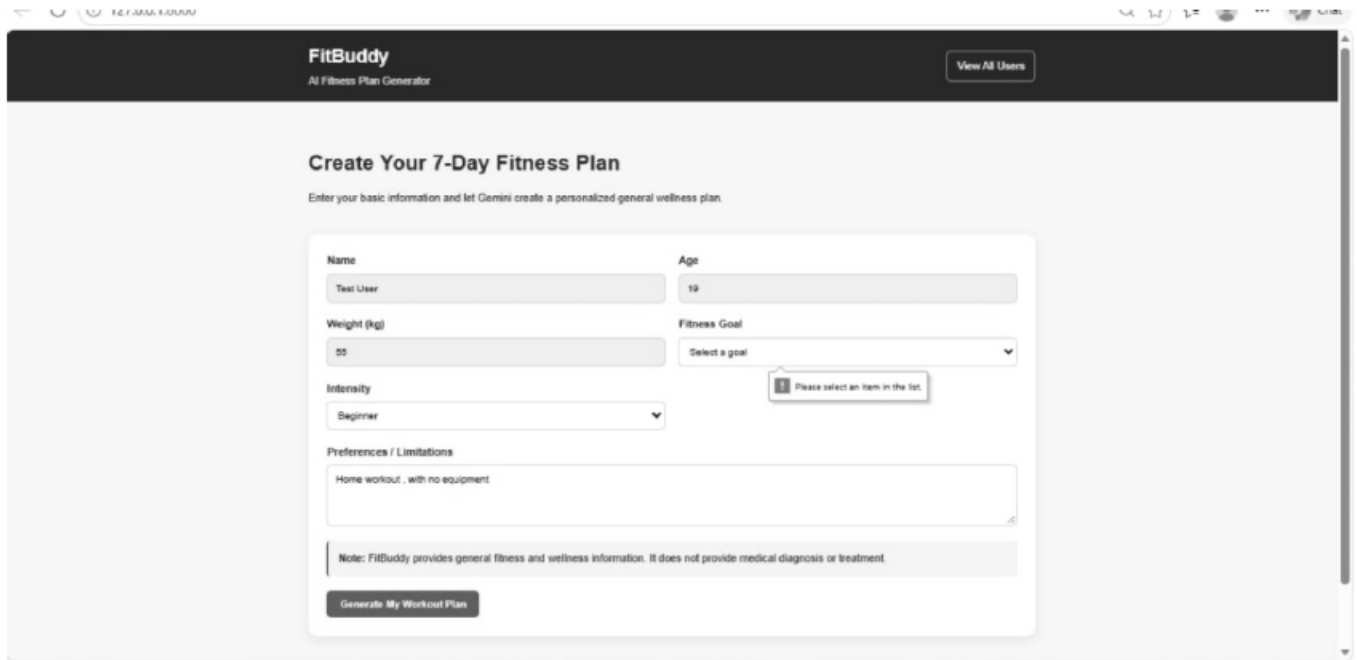
6.7 Testing Conclusion

The testing process covered the major functionalities of FitBuddy, including input validation, FastAPI endpoint testing, AI-based workout-plan generation, output display, Nutrition & Recovery information, feedback submission, updated-plan generation, and database storage. The screenshots provide application-level evidence for these testing activities.

Note: Coding screenshots are not included in Phase 6 because source-code evidence is covered under Phase 5/Development. Phase 6 focuses on the application working and the corresponding testing evidence.

6.4.8 Additional User Input Validation Evidence

The following screenshot shows the FitBuddy fitness-plan input page during validation testing. The required Fitness Goal field was left unselected, and the application displayed a validation message. This provides additional evidence for TC03 (required-field validation).



The screenshot displays the FitBuddy web application interface for creating a 7-day fitness plan. The header includes the 'FitBuddy' logo and the tagline 'AI Fitness Plan Generator', along with a 'View All Users' button. The main heading is 'Create Your 7-Day Fitness Plan', followed by the instruction 'Enter your basic information and let Gemini create a personalized general wellness plan.' The form contains several input fields: 'Name' (filled with 'Test User'), 'Age' (filled with '19'), 'Weight (kg)' (filled with '55'), 'Intensity' (a dropdown menu set to 'Beginner'), and 'Fitness Goal' (a dropdown menu with the text 'Select a goal'). A validation message, 'Please select an item in the list.', is displayed next to the 'Fitness Goal' field. Below these fields is a 'Preferences / Limitations' text area containing the text 'Home workout, with no equipment'. At the bottom of the form, there is a note: 'Note: FitBuddy provides general fitness and wellness information. It does not provide medical diagnosis or treatment.' and a 'Generate My Workout Plan' button.

Figure 8 – FitBuddy User Input Validation / Required Field Message

Testing Result: Validation message displayed successfully. Status: Pass.